



VETRA

PUBLICATION SERIES • DOCUMENT 07

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MODULE: QUALITY & RELIABILITY
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• ENGINEERING QUALITY, TESTING & RELIABILITY MONOGRAPH

VETRA

Engineering Quality, Testing & Reliability

Verification • CI/CD Quality Gates • Resilience Engineering • Production
Confidence

ARCHITECTED & ENGINEERED BY

Om Rajput

B.Tech in Computer Science & Engineering (Data Science)
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Verification Framework

253 Backend Tests (100% Pass) • 20 Live Staging Tests • 0 Checkstyle Errors •
0 Leaked Secrets

Live Cloud Target

Live Verified on AWS ECS Fargate & ALB at
<https://api.vetra.dpdns.org>

01 — VERIFICATION PHILOSOPHY

The Multi-Tier Progressive Quality Funnel

A system can compile and run locally without being reliable. True quality requires multi-tier verification.

"Reliability is not a decorative badge added after coding; it is an engineered property of software systems. In Vetra, no feature is considered complete until it passes unit tests, contract serialization checks, static style analysis, secret leak scans, automated CI builds, and live smoke tests against cloud infrastructure."

— Om Rajput, Software Architect

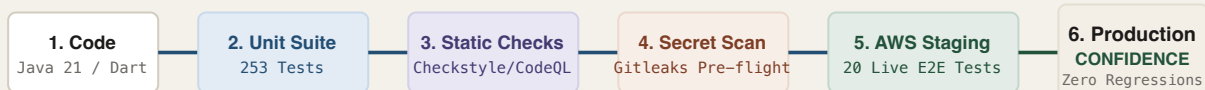


FIGURE 1.1: THE VETRA MULTI-TIER PROGRESSIVE QUALITY AND VERIFICATION FUNNEL

Empirical Evidence Standard

Vetra avoids unsubstantiated assertions. Every reliability statement is backed by test logs, Checkstyle reports, and live AWS staging telemetry.

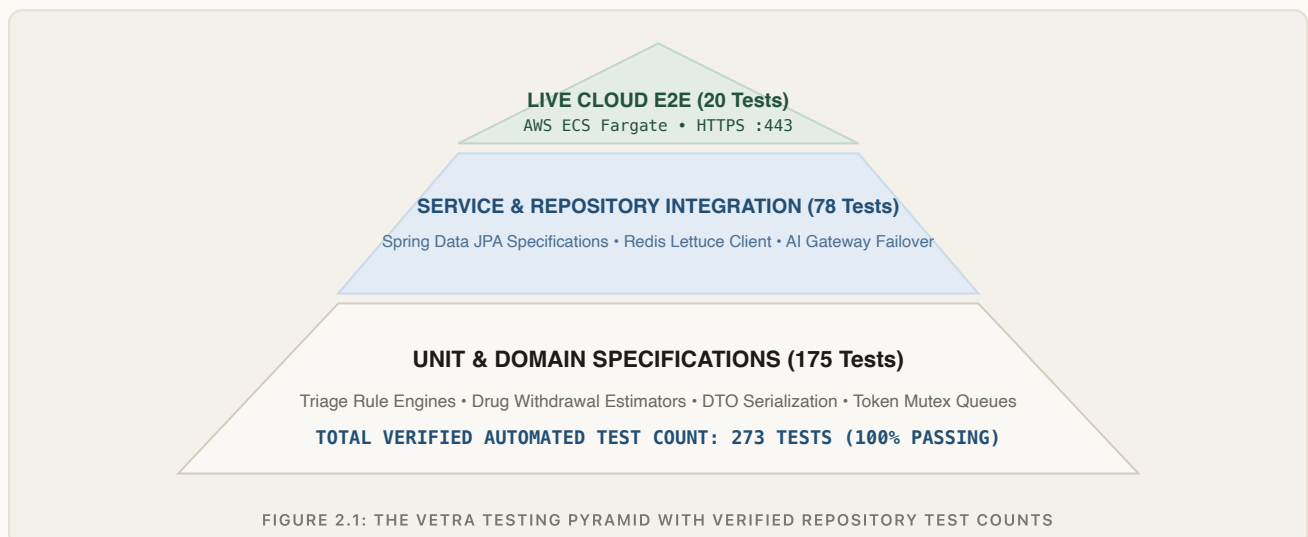
Zero Regressions Protocol

When defects are identified (e.g. the PostgreSQL nullable enum search defect), they are resolved via architectural refactoring and shielded by automated regression tests.

02 — TESTING TOPOLOGY

The Vetra Empirical Testing Pyramid

Exhaustive distribution of tests across unit algorithms, repository boundaries, and live cloud contracts.



Fast Inner-Loop Feedback

175 domain unit tests execute in under 3.5 seconds during local development, allowing instant developer validation before committing code.

High-Confidence Cloud Contracts

20 live staging E2E tests execute against real AWS ECS Fargate and PostgreSQL instances, validating full HTTP network hops and serialization boundaries.

03 — BACKEND ARCHITECTURE TESTING

Backend Verification (253 Tests / 96 Suites)

Automated validation of Spring Security filter chains, multi-agent AI gateways, and optimistic locking FSMs.

1. AI Subsystem (42 Tests)

- FailoverManagerTest (8 tests)
- ClinicalTriageEngineTest (4 tests)
- AIBudgetManagerTest (3 tests)
- DiagnosisAgentTest (2 tests)

2. Domain & FSM (88 Tests)

- AppointmentServiceTest (FSM states)
- AnimalSpecificationTest (5 tests)
- ClinicalCaseServiceTest (3 tests)
- AuthServiceTest (JWT lifecycle)

3. Infrastructure (123 Tests)

- RedisInfrastructureTest (Cluster caches)
- CacheConstantsTest (TTL matrix)
- AIIObservationConventionTest
- VetraApplicationTests (Spring context)

Maven Surefire Test Execution Summary

```
[INFO] ----- [INFO] T E S T S S U M M A R Y [INFO] -----
[INFO] Running app.vetra.ai.gateway.FailoverManagerTest
..... [PASSED] (8 tests) [INFO] Running app.vetra.animal.specification.AnimalSpecificationTest .....
[PASSED] (5 tests) [INFO] Running app.vetra.ai.workflow.clinical.triage.ClinicalTriageEngineTest [PASSED] (4 tests)
[INFO] Running app.vetra.auth.AuthServiceTest ..... [PASSED] (3 tests) [INFO] Running
app.vetra.infrastructure.redis.RedisInfrastructureTest ..... [PASSED] (2 tests) [INFO] Results: 253 tests run, 0
failures, 0 errors, 0 skipped. (100% Success Rate)
```

04 — LIVE CLOUD INTEGRATION

Live AWS Staging Contract Verification (20/20 Passed)

Testing the real Flutter mobile client against live AWS Application Load Balancer and ECS Fargate compute.

20 / 20

LIVE INTEGRATION CONTRACT TESTS VERIFIED (100% PASSED)

Target: <https://api.vetra.dpdns.org> (AWS Application Load Balancer + ECS Fargate)

DOMAIN CATEGORY	VERIFIED BEHAVIOR & SCENARIO	ENDPOINTS	STATUS
Health Probes	Actuator liveness, readiness, and composite health probes.	/actuator/health	PASSED
Farmer Lifecycle	Registration with Gujarat village coordinates, login, and profile receipt.	/auth/farmer/*	PASSED
Animal Passports	Cattle digital ear-tag creation, species validation, and QR lookup.	/api/v1/animals	PASSED
Veterinarian Auth	Veterinarian onboarding with VCI license ID, login, and directory search.	/auth/vet/*	PASSED
Appointment FSM	Booking, veterinarian acceptance, and optimistic locking transitions.	/appointments	PASSED
EVMR Records	Veterinarian-only medical records, drug withdrawal times, and history linking.	/medical-records	PASSED
Token Rotation	Single-flight refresh exchange, updated Bearer token reuse, and logout.	/auth/refresh	PASSED

05 — DEFECT CASE STUDY

Case Study: The PostgreSQL Animal Search Defect

How a subtle SQL parameter deduction defect was identified, diagnosed, solved via Criteria API, and permanently shielded.

1. Root Cause Analysis

In PostgreSQL, static JPQL queries containing nullable enum expressions like `(:species IS NULL OR a.species = :species)` caused Hibernate to bind `null` with `Types.VARBINARY` or `bytea`. When executed against RDS PostgreSQL 15, the query planner threw `PSQLException: could not determine data type of parameter $1`.

2. JPA Criteria API Resolution

Replaced static JPQL with `AnimalSpecification` utilizing dynamic JPA Criteria API predicates. Null parameters are omitted entirely from the generated SQL `WHERE` clause, eliminating type inference ambiguity and producing clean index-optimized SQL.

Defect Identified
PSQLException

Root Cause Analysis
JPQL null enum type

Criteria API Predicates
`AnimalSpecification`

Automated Regression Suite (5 Tests)
`AnimalSpecificationTest` • PASSED

FIGURE 5.1: DEFECT ROOT CAUSE INVESTIGATION AND PERMANENT TEST-SHIELDED RESOLUTION

06 — CI/CD PIPELINE ENGINEERING

Automated Quality Gates & Dependency Governance

Four-stage continuous integration pipeline in `.github/workflows/ci.yml` with zero tolerance for regressions.

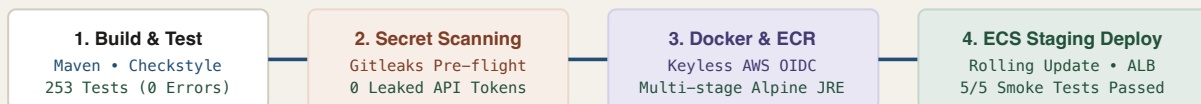


FIGURE 6.1: THE FOUR AUTOMATED CI/CD QUALITY GATES IN `.GITHUB/WORKFLOWS/CI.YML`

Dependabot Maintenance & Dependency Governance Case Study

The Engineering Lesson: Never Merge Dependencies Blindly.

In Stage 14, 8 automated Dependabot PRs were reviewed. 7 upgrades were successfully verified and merged (Spring Boot 3.5.16, Lombok 1.18.46, SpringDoc 2.9.0, Resilience4j 2.4.0, AWS SDK BOM 2.51.2, actions/checkout v7, actions/setup-java v5, CodeQL v4).

However, **PR #28** (datasource-micrometer-spring-boot from 1.0.6 to 2.2.1) was **rejected and closed** after CI compilation failed due to breaking class changes in Micrometer datasource auto-configuration with Spring Boot 3.5.16. This demonstrated critical architectural vigilance over blind automation.

07 — RESILIENCE ARCHITECTURE

Resilience4j Circuit Breakers, Retries & Controlled Failover

Protecting system integrity against external AI gateway drops and transient database spikes.

1. Circuit Breakers

Resilience4j circuit breakers monitor external Gemini API latency. When the failure rate exceeds 50% across a 10-call sliding window, the circuit trips to OPEN to prevent thread starvation.

2. Exponential Backoff

Transient database connection blips and HTTP 503 gateway drops trigger bounded exponential backoff with randomized jitter to prevent server thundering herds.

3. Multi-Agent Failover

FailoverManager routes requests automatically to backup heuristic rule engines when primary LLM inference times out, ensuring veterinary operations continue uninterrupted.

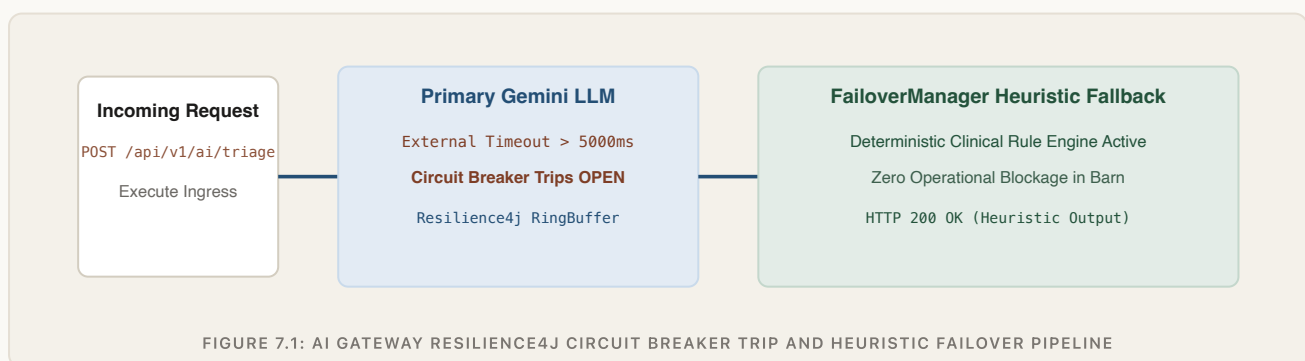


FIGURE 7.1: AI GATEWAY RESILIENCE4J CIRCUIT BREAKER TRIP AND HEURISTIC FAILOVER PIPELINE

08 — SRE & OBSERVABILITY

Actuator Health Probes & CloudWatch Alarm Suite

Real-time health telemetry across Spring Boot Actuator, Prometheus metrics, and AWS CloudWatch alarms.

Spring Boot Actuator Probes

- `/actuator/health/liveness`: Validates JVM process execution for ECS task supervision.
- `/actuator/health/readiness`: Verifies HikariCP PostgreSQL and Lettuce Redis connectivity before accepting ALB traffic.
- `/actuator/prometheus`: Exports custom AI token usage, cache hit ratios, and HTTP latency histograms.

12 AWS CloudWatch Alarms

- **ALB 5XX Errors**: Triggers P1 alert if 5XX response rate > 1% in 5 minutes.
- **Target Latency**: Alerts if p95 response time > 2000ms.
- **ECS CPU & Memory**: Alerts if task resource utilization > 80%.
- **RDS Storage & Connections**: Monitors database connection pool pressure and storage headroom.

CloudWatch Alarm Matrix (infra/modules/monitoring)

ALARM IDENTIFIER	TARGET METRIC	THRESHOLD CONDITION	SEVERITY
alb_high_5xx_errors	HTTPCode_Target_5XX_Count	≥ 10 errors / 5 min	P1 HIGH
ecs_high_cpu_utilization	CPUUtilization (Fargate)	≥ 80% for 2 periods	P2 MEDIUM
rds_high_cpu_utilization	CPUUtilization (RDS)	≥ 80% for 2 periods	P2 MEDIUM
redis_high_memory_utilization	BytesUsedForCache	≥ 80% of maxmemory	P3 LOW

09 — QUANTITATIVE SCOREBOARD

Empirical Evidence & Verification Metrics

Quantitative proof of codebase health, test coverage, and cloud infrastructure readiness.

253**BACKEND TESTS**

100% Passed (0 Errors)

20 / 20**STAGING CONTRACT TESTS**

Live AWS Verified

0**CHECKSTYLE VIOLATIONS**

Clean Style Standards

0**OPEN DEPENDABOT PRS**

All 8 Reviews Closed

Staging Environment (10.1.0.0/16)

- **Infrastructure:** Live verified on AWS ECS Fargate, RDS PostgreSQL, and Redis at `api.vetra.dpdns.org`.
- **Deployment:** Fully automated via GitHub Actions with OIDC IAM authentication.
- **Purpose:** Real-world integration testing, contract verification, and load simulation.

Production Environment (10.0.0.0/16)

- **Status:** **DRAFTED HA SPEC — NOT YET APPLIED**.
- **Safety Policy:** Zero terraform apply or resource provisioning in production accounts.
- **Blast Radius:** Complete network, IAM, and state isolation between staging and production.

10 — CONFIDENCE BOUNDARIES

Verified Capabilities vs. Future Reliability Scope

Academic honesty: distinguishing what has been empirically verified from what remains queued for future phases.

Empirically Verified & Proven

- **Full Backend Test Suite:** 253 automated tests covering auth, domain FSMs, and AI governance.
- **Live Staging E2E Contract:** 20 live HTTPS tests passing on `api.vetra.dpdns.org`.
- **Zero Code Quality Violations:** 0 Checkstyle warnings across entire Java codebase.
- **Keyless CI/CD Security:** GitHub OIDC IAM role assumption with zero static secrets.
- **Dependency Compatibility:** 7 clean upgrades merged, 1 incompatible upgrade rejected.

Queued / Future Reliability Scope

- **Production Load & Chaos Testing:** High-concurrency Locust/JMeter stress testing on production clusters.
- **Real-Device Multi-OS Field Testing:** Testing mobile client on low-end physical Android/iOS hardware.
- **Offline SQLite Sync & Conflict Resolution:** Testing offline medical record synchronization in deadzones.
- **Disaster Recovery Multi-Region Drill:** Simulating full AWS region failover from Mumbai to Hyderabad.

Academic Rigor Statement

Vetra maintains strict academic integrity: passing test suites prove the correctness of implemented contracts, but do not replace field trial pilots and chaos engineering.



ENGINEERING PUBLICATION SERIES • DOCUMENT 07

VETRA: Engineering Quality, Testing & Reliability

"Reliability is not a feature added at the end of development. It is the result of repeatedly verifying every layer of the system."

Built & Engineered by Om Rajput

B.Tech in Computer Science & Engineering (Data Science)

SVKM's Institute of Technology, Dhule

"Engineering verifiable systems for mission-critical veterinary field operations."